

Machine Learning Model Report

Fraudulent Transactions

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I. Introduction

The goal of our task is to create a machine learning model capable of predicting whether a given bank transaction is fraudulent. To achieve this, we are given a dataset consisting of a total of 227,845 entries, each representing a bank transaction. Every transaction is labeled as either genuine (0) or fraudulent (1). For this reason, the problem at hand is a binary classification problem. Each entry consists of a total of 30 features, where the first is a timestamp and the last is the amount of the transaction. The remaining 28 features are unknown numerical values. Overall, the dataset is highly unbalanced, with only 394 entries (about 0.17%) labeled as fraudulent.

II. Implementation / ML Process

A. Models

Since this task is a binary classification problem, a logistic regression model is perfectly suited, as it is, by default, a binary classifier that outputs values between 0 and 1, modeling the probability of an entry being fraudulent.

Our second choice, besides logistic regression, is a decision tree, as the numerical values of our dataset are well suited for the split mechanism at each non-leaf node. Additionally, just like logistic regression, decision trees are also, by default, well suited for binary classification.

B. Preprocessing

For both models, we decided on a standard training-test split ratio of 80/20.

Since the occurrence of fraudulent transactions was extremely low compared to non-fraudulent transactions, we implemented simple oversampling with replacement to increase the number of fraudulent entries while trying to preserve the majority class distribution. To find a good ratio, we manually tried out different values

for the oversampling factor on the default models provided by scikit-learn.

Based on the performance of the two models, we settled on a fraud ratio of 2.6% for the logistic regression classifier and 1% for the decision tree classifier.

C. Hyperparameters

In logistic regression, adjustable hyperparameters include the C value for the regularization strength, the form of penalty (L1 / L2), the scaler (StandardScaler / RobustScaler) used to center the data before evaluation, and the solver (liblinear / saga) used by the model. To find out which values work best, we executed a grid search that evaluates different settings for our hyperparameters using 5-fold cross-validation. After trying out different combinations, we settled on C:

'0.01', penalty: 'l2', solver: 'liblinear', and the RobustScaler.

Hyperparameters concerning our decision tree include the criterion for the impurity measure

(gini / entropy / log _loss), min _samples split (minimum number of samples required to split an internal node), and min samples leaf (minimum number of samples required to be a leaf node). Once again, we executed a grid search evaluation with 5-fold cross-validation to find the optimal values for our hyperparameters. These optimal values were: criterion: 'gini', min samples split: '2', min _samples leaf: '10', and the StandardScaler.

For both models, we wanted to optimize the recall score during the grid search, in order to minimize false-negative predictions while still trying to keep false-positive predictions as low as possible.

III. Results

Generally speaking, both of our models performed quite well on the training and validation sets when looking at the precision and ROC-AUC scores.

A. Logistic Regression

Logistic Regression	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Training Set	0.9992	0.7393	0.8190	0.7771	0.9822
Validation Set	0.9993	0.7647	0.8228	0.7927	0.9815

Table 1: Comparison of performance for the Logistic Regression classifier given the test and validation set

As seen in Table 1, the logistic regression classifier performed very differently when evaluated across distinct metrics. While the accuracy and ROC-AUC scores are very high, the other metrics, in comparison, turned out to be rather mediocre.

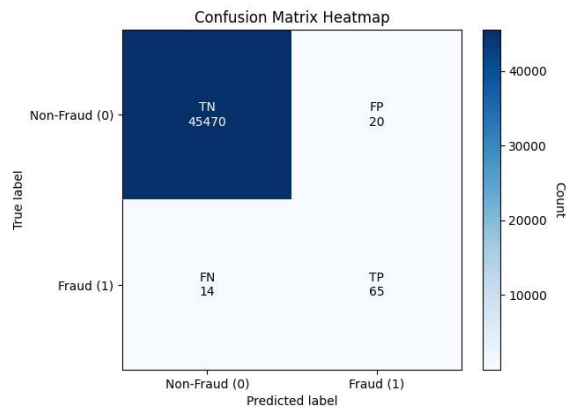


Figure 1: Confusion-Matrix for Logistic Regression on the validation set

In Figure 1, one can see the balance we aimed to achieve between false-negative (FN) and falsepositive (FP) predictions. When trying to minimize the false negatives, the number of falsepositive predictions appeared to increase exponentially.

B. Decision Tree

Decision Tree	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Training Set	0.9994	0.7664	0.9270	0.8391	0.9999
Validation Set	0.9990	0.6923	0.7975	0.7412	0.9111

Table 2: Comparison of performance for the Decision Tree classifier given the test and validation set

Similar to the logistic regression classifier, the performance differs significantly depending on the metrics used. However, in contrast to logistic regression, when comparing the performances between the test and validation sets, one can see that the decision tree classifier performs worse on all metrics when evaluated on the validation set.

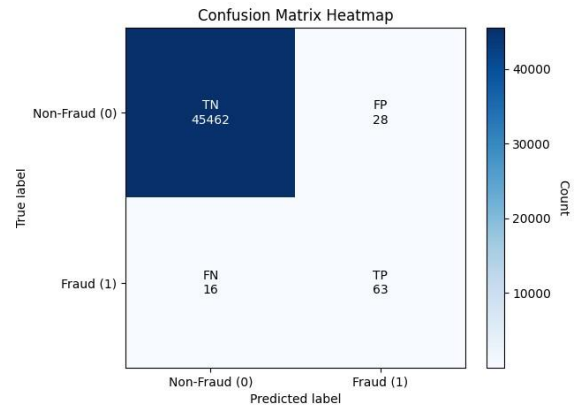


Figure 2: Confusion-Matrix for Decision Tree on the validation set

IV. Discussion

The logistic regression classifier performs well on the ROC-AUC metric and appears to generalize very well for unseen data. Unfortunately, the same cannot be said about the decision tree classifier, which evidently has a major issue with generalization and performed significantly worse on all metrics when evaluated on the validation set. Although overfitting is very typical for decision tree classifiers, we believe that it would be possible to achieve better results if the dataset contained more fraudulent samples.

The small percentage of fraudulent samples was probably the biggest issue with this problem, but it could perhaps have been further mitigated if we had used synthetic oversampling—meaning that we do not sample from the original dataset, but rather generate new fraudulent samples based on all existing fraudulent samples.

We also attempted to use dimensionality reduction but rather quickly came to the conclusion that, due to the simple nature of the dataset, we could not derive any added value from this method.

V. Conclusion

On the test set provided, we achieved a ROC-AUC score of 0.9850 for our Logistic Regression classifier, and a score of 0.8925 for our Decision Tree classifier.

This further strengthens our opinion, that the Decision Tree classifier has overfit the training data to much. However we are satisfied with the performance of our Logistic Regression classifier and in retrospect would not choose a Decision Tree classifier for this task again but rather a

Support-Vector-Machine or even a Neural Network.